

An field experiment was carried out under the Department of Plant Pathology, Hajee Mohammad Danesh Science & Technology University, Dinajpur during the period of July 2009 to December 2009 to find out the fungicidal management of sheath blight of rice. Three fungicides viz. Bavistin 50 WP, Tilt 250 EC and Folicour EW 250 were sprayed against *R. solani* causing sheath blight on three aman rice varieties viz. Hybrid Hera, BRRI Dhan-49 and BR-11. Control plot was sprayed with plain water. The experiments were laid out in two factors Randomized Complete Block Design (RCBD) with three replications. From the present findings it may be concluded that Folicour EW 250 was a promising fungicide for lowering sheath blight incidence and BR-11 was the best variety among the three rice varieties in aspect of yield.



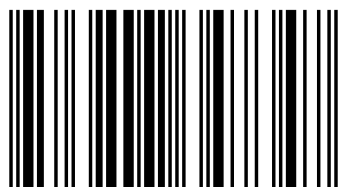
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Fungicidal Management of Sheath Blight of Rice

Sheath Blight Management



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ABSTRACT

An field experiment was carried out under the Department of Plant Pathology, Hajee Mohammad Danesh Science & Technology University, Dinajpur during the period of July 2009 to December 2009 to find out the prevalence and fungicidal management of sheath blight of rice. Three fungicides viz. Bavistin 50 WP, Tilt 250 EC and Folicour EW 250 were sprayed against *R. solani* causing sheath blight on three aman rice varieties viz. Hybrid Hera, BRRI Dhan-49 and BR-11. Control plot was sprayed with plain water. The experiments were laid out in two factors Randomized Complete Block Design (RCBD) with three replications. The tallest plant height (116.80 cm) was recorded in Hybrid Hera variety and the shortest plants height (106.00 cm) was obtained from the BR-11 variety. The percent relative lesion height (% RLH) was found significantly in deferent with some extent. Comparatively, the lower % RLH (3.42) was recorded in Folicour EW 250 sprayed plot where the highest % RLH (4.16) was found in control plot. The percent decrease of relative lesion height (17.78%) was recorded in the Folicour EW 250 sprayed plot over control. Highest numbers of infected tiller/hill (5.28) were recorded in the control plot where the lowest numbers of infected tiller/hill (4.92) was recorded in Folicour EW 250 sprayed plot but incase of varieties the highest numbers of infected tiller/hill (8.31) was recorded in BR-11 variety and the lowest numbers of infected tiller/hill (3.31) was recorded in Hybrid Hera. The highest yield (t/ha) (5.81) was recorded in Folicour EW 250 sprayed plot and the lowest yield (t/ha) (4.24) was recorded in the control plot. The fungicides Folicour EW 250 increased yield (37.02%) than over control. In case of varieties, the highest yields (5.93 t/ha) was recorded in Variety BR-11 and the lowest yield (5.10 t/ha) was obtained from Hybrid Hera variety. From the present findings it may be concluded that Folicour EW 250 was a promising fungicide for lowering sheath blight incidence and BR-11 was the best variety among the three aman varieties in aspect of yield.

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CHAPTER 1

INTRODUCTION

Rice (*Oryza sativa* L.) constitutes the staple food of 60% of the world population; rice is also the most important cereal crops in Bangladesh. It occupies about 75% of the total cropped area covering 26.1 million acres of land in Bangladesh. In Bangladesh, about 11.32 million hectare cropped areas are under rice cultivation and produced 315.08 lac metric tons grain during the period of 2008-2009 (Anonymous, 2009). Bangladesh receives an average yield of rice is 3.8 t/ha that is lower compared to China (5.3 t/ha), Indonesia (4.36 t/ha) and Vietnam (4.72 t/ha) (Anonymous, 2008) .

There are so many constraints responsible for low yield of rice in Bangladesh. Among them diseases are considered to be the most important one. Thirty six fungal, twenty-one viral, bacterial and six nematode diseases are recorded in rice (Ou, 1985). So far, in Bangladesh 31 rice diseases have been identified of which ten are considered as major (Miah and Shahjahan 1987). Among them five diseases, viz. tungro, bacterial blight, sheath blight, blast and ufra have been considered as most important because of their widespread occurrence and significant damage potential.

Rice sheath blight prevalent in almost all rice growing areas and in all season of Bangladesh and is one of the major constraints to rice production in the country (Miah *et al.* 1985). The disease first reported in Bangladesh in 1973 (Miah *et al.*

1995). Both local, high yielding and hybrid varieties are susceptible to the disease. The intensity of the disease is severe in Aus and T. Aman seasons. Northern and central part of the country experiences frequent epidemic outbreak of the disease.

Sheath blight has becoming as an increasing problem for rice cultivation in Bangladesh. High temperature and high relative humidity during crop growth favour development of sheath blight disease. Short stature, high tillering and high nitrogen loving varieties are comparatively more susceptible as the micro climates inside the rice canopy is more favorable than those of the traditional ones of tall plant type with low tillering ability (Miah *et al.* 1985). This disease is a major constraint in getting potential yield from the modern varieties. The yield reduction due to Sheath blight in Bangladesh has estimated to be from 14 to 31% under experimental and farmers field condition (Shahjahan *et al.* 1986).

No cultivars or genotypes have so far been found to possess resistance to sheath blight disease (Sharma *et al.* 1985 and Ou. 1985). Therefore, emphasis should be paid on other management options. Control of sheath blight by fungicides has been attempted for many years in different countries of world.

Present disease management practices concentrate on foliar fungicides, crop rotation and manipulation of cultural practices. However, culture manipulations (Shahjahan *et al.* 1990) crop rotation (Damicone *et al.* 1993) and biological control schemes (Sneh *et al.* 1996) appear to be limited effect. To date breeding for sheath

blight disease resistance is not vary successful due to lack of sources of immunity or of high level of resistance (Rao 1995, Hashiba and Kobayashi 1996).

Globally, chemical control represents the most significant practice to reduce yield loss caused by sheath blight (Sarkar *et al.* 1991 and Acharya *et al.* 1997). In Bangladesh, Propiconazol (Tilt 250 EC) and Carbendiazim (Bavistin) are being used. However, their efficacy appeared to be poor and highly variable from one geographic location to another and profitability has always been questionable.

Thus, on the basis of above facts, thoughts and views the present study has been aimed to achieve the following objectives:

- ❖ To find out sheath blight incidence of three aman rice varieties viz. Hybrid Hera, BRRI Dhan-49 and BR-11.
- ❖ To evaluate the efficacy of three fungicides against sheath blight of rice.

CHAPTER 2

REVIEW OF LITERATURE

Sheath blight (*Rhizoctonia solani*) is a worldwide important disease of rice. Many works and researchers have been trying to management of sheath blight of rice with different tactics. For better and precise presentation only the related literatures are presented here.

2.1. History and Distribution

The occurrence of sheath blight in rice caused by *Rhizoctonia solani* Kuhn was first described by Miyake (1910) in Japan and subsequently its occurrence on rice has been reported from almost all rice growing countries of world in Asia, Africa and Americas (Kozaka, 1975; Ou, 1985; Shahjahan *et al.*, 1987).

Rice sheath blight disease caused by the fungus *Thanetophorus cucumeris* (Frank) Donk (*Rhizoctonia solani* Kuhn) was first reported in Bangladesh by Talukdar in 1968. Later on, Miah (1973) confirmed its occurrence and reported that the incidence of the disease was increasing in Bangladesh. With the intensification of rice cultivation, it has now become a major disease of rice in the country.

Sugiyama (1988) Reported that the areas by sheath blight in Japan increased from 35% of the cultivated areas in the 1960's to 46% in 1970's and 48% in 1980's.

Suryadi *et al.* (1991) reported that the areas affected by sheath blight in Japan increased from 35% of cultivated areas in the 1960's to 46% in 1970's and to 48% in 1980's.

2.2. Symptomatology:

Sheath blight causes spots on the leaf sheath. The spots are at first ellipsoid or ovoid, somewhat irregular, greenish-grey, and varying from 1 to 3 cm long. Sclerotia are formed on or near these spots, but are easily detached. The size and colour of spots and the formation of sclerotia depend upon environmental conditions mycelium of the fungus may grow over the surface of the leaf sheaths and can spread to a considerable distance (Ou, 1985).

Singh *et al.* (1988) reported that the pathogen normally attacked the leaf sheath and leaf blade but the symptoms were also found on emerging panicles which chaffy, grayish brown and matted together by fungal mycelium. Numerous white and brown sclerotia were found on diseased panicles.

Rangaswami and Mahadevan (1999) reported that the pathogen causes spots or lesion mostly on the leaf sheath, extending to the leaf blades under favourable conditions. At first, the spots are greenish gray, ellipsoid or ovoid, and about 1 cm long. They enlarge and may reach 2 or 3 cm in length and become grayish white with brown margins and somewhat irregular in outline. In the advance stage brown sclerotia are formed which are easily detached from these spots. Under

humid conditions, the fungal mycelium spreads to other leaf sheaths and blades. Eventually, the whole sheath rots and affected leaf can easily be pulled off from the plant. In severe case all usually attacked at the tillering stage, when the leaf sheaths become discolored at or above water level.

Rush and Lee (1992) reported that the symptoms of sheath blight do not appear until plants are in the late tillering or early internode elongation growth stages. Initial symptoms consist of circular, oblong or ellipsoid, green-grey, water-soaked spots about 1 cm long that occur on the leaf sheath near the waterline. The lesions enlarge to approximately 2-3 cm in length and 1 cm in width, and the centers of the lesions become pale green or white and are surrounded by an irregular purple-brown border. Under favourable conditions i.e., low sunlight, humidity near 95% and high temperature (28-32⁰C), infection spreads rapidly by means of runner hyphae to upper plant parts including leaf blades, and to adjacent plants. Lesion on the upper portion of plants may coalesce to encompass entire leaf sheaths and stems. Sclerotia are produced superficially on or near infected tissue after symptoms first appear and are loosely attached and easily dislodge from the plant when they are mature. Heavily infected plants produce poorly filled grains, particularly in the lower portion of panicle. Additional losses in yield result from increased lodging or reduced ratoon production as a result of death of culm.

Yu, C. M. (1975) reported that outer leaf sheaths are first affected, gradually extending towards the inner sheath. Sheath blight lesion spread more rapidly on

the leaf sheath blade than the sheath, and spread is hastened by loosening. Lesion length increases faster at heading than at booting and maximum at tillering stage. At booting the developments is more rapid on lower leaf sheaths than the upper, but reverse at heading of the rice plants.

2.3. Factors responsible for the growth of *Rhizoctonia solani* and sheath blight

Sheath blight is basically a disease of wet conditions. It is especially destructive under conditions of high humidity and high temperature. Temperature near the leaf sheath will be related to air temperature but humidity within the crop stand will vary according to tillering nature of the variety, density of stand and amount of fertilizers added. Therefore, very close stand and heavy doses of nitrogen, especially when the crop has passed tillering stage, pre-dispose the plants to sheath blight. Optimum temperature for growth and sclerotial formation by the fungus is about 30⁰ C, the minimum being 10⁰ C and maximum 40-42⁰ C. A sudden rise and fall in temperature accelerates sclerotial formation. Infection occur in the temperature range of 25-35⁰ C but optimum conditions are temperature of 30⁰ C and 32⁰ C and relative humidity 96 to 97%.

The pure cultures of *R. solani* isolates from high temperature regions grow well on PDA at 35⁰ C but tended to grow poorly at 12⁰ C, whereas those from low temperature regions at 35⁰ C and well at 12⁰ C reported by Hashiba *et al.* (1974).

There was no difference in growth between the isolates at 15⁰ C. With a daily average temperature of 26-28⁰ C (daily maximum 30-32⁰ C) the rate of upward development of the disease caused by higher temperature isolates was greater than that of these caused by the low temperature isolates.

There was no significant difference in the growth of *R. solani* on rice leaf sheath held in the dark or light but lesions development more rapidly in the light reported by Yuno *et al.* (1978). Mycelial growth increased with temperature 20-28⁰ C. From 25-26⁰ C hrs. after disease development, lesions of leaf sheaths were cleanly defined and the lesions diameter then remained constant.

Leano (1993) reported that the infection efficiency in sheath blight significantly increased with prolonged wet and dry daily cycles, particularly with three (3) days of intermittent wet and dry (12/12 hrs.) periods.

Soil moisture was important factor determining the survival and persistence of the fungal pathogen in soil reported by Kannaiyan and Prasad (1979). The increase in moisture levels gradually decreased the seedling infection. Low moisture favoured the survival of *R. solani*, whereas, higher moisture inhibited the survival and the multiplication of the propagules of *R. solani*. Higher moisture content also stimulated the bacterial activity which in turn might result in lysis of *R. solani* mycelium.

Leano (1993) observed that sheath blight intensity and yield loss were higher in plots where flooding was intermittent compared to continuously flooded ones.

Lakpale *et al.* (1994) observed that viability of sclerotia remained unchanged when sclerotia were submerged for periods up to 96 hrs. followed by drying treatments of 24-168 hrs. However, viability decreased with an increase in submergence until it was completely lost after 168 hrs. Submergence also lowered the inoculum potential by reducing the production of sclerotia from those which had been under water.

2.4. Plant growth stages and sheath blight severity

Boyette and Lee (1979) reported that plant growth stages have significant effects on *R. solani* Kuhn and development of rice sheath blight in the field. In field trials maximum yield losses occurred when plant becomes infected at the $\frac{1}{2}$ in internode elongated growth stage.

Kim and Min (1983) reported that the percentage of infected hills/stems increased more rapidly from 11 July to 1" August than 1" September (i. e., Booting stage than flowering stage of crop growth).

Miah *et al.* (1985) reported that the disease severity and yield losses were significantly higher when inoculated at booting stage than when inoculated at maximum tillering, panicle initiation and flowering stages.

Suparyono and Nuryanto (1991) observed the severity of sheath blight and its effect on rice yield depended on the rice variety and the host growth stage at which plants were inoculated. Disease development was greatest when plants at maximum tillering and primordial stages were inoculated with *Rhizoctonia solani*. Disease severity of the early maturity varieties was less when plants were inoculated at the later stage, although disease severity was affected long duration rice varieties. Yields of both early and long duration varieties were significantly reduced by the disease when the plants were inoculated at the maximum tillering and primordial plant growth stages. Yield reduction decreased when plants of the early maturing varieties were inoculated at the later stage, although the reduction of the yield loss was smaller on the long duration varieties.

Sharma and Teng (1996) studied the development of sheath blight on two rice cultivars IR-42 and IR-72 inoculated at different growth stages with *Rhizoctonia solani*. They reported that disease development was highest at later growth stages. Disease progress was faster at flowering and booting stages than at tillering and panicle initiation stages in both cultivars. The percent productive tillers, grain weight per plants, filled grain number per panicle, 1000-grain weight, filled grain percent and total biomass per plant were usually higher at early stages of inoculation and lower at booting and flowering stages of inoculation. The disease development was also higher at later growth stages and consequently the yield parameters also declined. Yield losses at different stages of inoculation ranged from 23.01 to 32.15% and 13.53

to 34.83% in IR-42 and IR-72, respectively. The yield losses were higher in flowering stage than those of other three (3) stages in both the cultivars.

2.5. Management of Sheath Blight with Chemicals

Arunayanart *et al.* (1986) tested 13 fungicides for sheath blight control where Pencycuron 25% WP, Validamycin 3% liquid, Carbendazim 60% WP, and Propiconazole 25% EC were most effective.

Dev and Mary (1986) tested seven fungicides to control sheath blight of rice. All the fungicides checked sheath blight intensity. Validamycin and carbendazim performed similarly and significantly better performance than other fungicides.

Jones *et al.* (1987) reported that propiconazole (Tilt) applied twice or propiconazole (Tilt) followed by benomyl significantly reduce disease severity and increase yield. An economic return from propiconazole (Tilt) could be anticipated when >5% diseased tillers were observed at the panicle differentiation growth stage.

Thangasamy and Rangaswamy (1989) tested control of sheath blight caused by *Thanatephorus cucumeris* by carbendazim and mancozeb applied at different crop growth and disease development stage. Fungicides were applied when disease severity was 1, 3 and 5 at panicle initiation (PI) or at heading stage. Carbendazim and mancozeb sprayed at PI and 15 day after PI controlled disease development.

Groth *et al.* (1993) evaluated a number of fungicides for control of rice disease. Tilt (Propiconazole) gave good sheath blight control but did not control blast. Rovral (Ipridion) was effective against sheath blight and gave good yield stability.

Parveen (1998) observed that the chemical's (Tilt) effect against sheath blight lesion development was found to be statistically significant but no significant difference on lesion length was obtained between single and double sprays of Tilt 250 EC. She conclude that sheath blight disease of rice could be controlled by only one spray of Tilt 250 EC.

Kandhari and Gupta (1999) reported that benomyl, carbendazim, ediphenfos and IBP have been extensively used for the control of sheath blight of rice in India. Two new organophosphorous compounds, G/F T-3 and G/F T-9 highly potent against *R. solani*

Mia (1999) recommended Tilt 250EC at the rate of 1L/ha for spraying at panicle initiation to booth stage for successful control of sheath blight disease of rice from Bangladesh Rice Research Institute. He also suggest Homai or Topsin M at the rate of 2 kg/ha at panicle initiation or boot stage.

Akter *et al.* (2001) conducted an experiment by using six fungicides. namely, Bavistin 50WP (carbendazim), Contaf 5EC (hexoconazole), Forastin 50WP

(carbendazim), Anvil 5SC (hexoconazole), Tilt 250EC (propiconazole), and Thiovit 80WP (micronized sulfur), and a fertilizer, muriate of potash, in Gazipur, Bangladesh, in 1999 and 2000 against sheath blight of rice (cv. Swarna) caused by *Rhizoctonia solani*. They found that Contaf appeared to be the best in reducing the percent relative lesion height, percent disease index (PDI), and tiller infection.

Hossain and Mia (2001) evaluated the effectiveness of four fungicides against sheath blight of rice where two foliar sprays with Tilt 250 EC @ 0.1% caused significant reduction in sheath blight severity and improved grain yield.

Hossain and Mia (2001) carried out an experiment in Rajshahi, Bangladesh, during T. aman season (June to December) of 1998 to evaluate the effectiveness of four fungicides. i.e. Aimcozim 50 WP (Carbendazim), Bavistin 50 WP (Carbendazim), Shincar 50WP (Carbendazim), and Tilt 250 EC (Propiconazole), and two additional rates of muriate of potash (MP), i.e. 20 and 40 kg/ha, against sheath blight (*Rhizoctonia oryzae*) of rice. The application of two additional 40 kg MP/ha in two equal splits as top dress and two foliar sprays with Tilt 250 EC at 0.1% caused significant reduction in sheath blight severity and improved grain yield.

Sitansu *et al.* (2001) evaluated the efficacy of Iprodione (Rovral 50) and carbendazim (Bavistin 50 WP) both at 0.5, 1.0, 2.0 or 4.0 ppm as single and combinations against rice sheath blight caused by *R. solani* and found the

single doses of carbendazim best in field condition but in vitro single dose of iprodione showed best.

Johnson *et al.* (2002) conducted an experiment with hexaconazole against *R. solani* causing rice sheath blight. They reported that hexaconazole did not remarkable change in peroxidase activity of *R. solani* inoculated rice plant.

Chahal *et al.* (2003) reported that Propiconazole (0.1%), edifenphos (0.1%), iprodione (0.3%), carboxin (0.2%) and carbendazim (0.1%) effectively controlled sheath blight of rice.

Biswas (2004) evaluated the relative efficacy of mix fungicides showing 250 EC (125 g epoxyconazole + 125 g carbendazim), Auintal 50 WP (25%) iprodione + 25% carbendazim) and armune 30 EC (15 g rpmiconazole + 15 g difenoconazole) against sheath blight (caused by *R. solani*) of rice.

Mian *et al.* (2004) conducted two field experiments in Bangladesh during T. Aus and T. Amon session in 2002 to find out the efficacy of eight fungicides viz. Forastin 50 WP (carbendazim), Genuine 50 WP (carbendazim), Cindazim 50WP (carbendazim), Akonazole 250 EC (propiconazole), Abgen 50%F (carbendazim), Goldazim 50WP (carbendazim), Folicur EW 250 (tebuconazole) and Tilt 250 EC (propiconazole) in controlling the sheath blight of rice (*Rhizoctonia solani*). All the fungicides were sprayed twice at 5 and 20 days after inoculation at recommended doses. Akonazole and Folicur appeared to be most effective in

reducing percent tiller infection, percent relative lesion height and percent disease index (PDI). In terms of PDI reduction over control, Akonazole and Folicur showed good performance in both sessions ranging from 59.96 to 80.29% and from 53.79 to 80.27%, respectively. The significant increase in grain yield was also recorded by applying Akonazole and Folicur over the untreated control.

Ashrafuzzaman *et al.* (2005) conducted an experiment to assess the single and combined doses of ash (3 t ha^{-1}), bleaching powder (10 ppm), poultry manure (5 t ha^{-1}) and bavistin (0.05%) for the management of sheath blight caused by *R. solani* Kuhn and their effects on yield components and yield of cv. BRRI dhan 39. They found that highest weight of thousand grains, grain yield per hill, straw yield per hill and lowest percent of unfilled grains were obtained when ash, bleaching powder, poultry manure and Bavistin used in combination.

Lore *et al.* (2005) conducted an experiment to evaluate different fungicides against sheath blight (*R. solani*) of rice under artificial epiphytotic conditions (K. purthala, Punjab, India, during 2002 and 2003) and indicated that monoceren (pencycuron) 250 SC exhibited superior efficiency over other fungicides such as RIL 010/F1 25SC, RIL 010/F1 50 SC, Rhizolex 50 WP, Rhizocin 3L, Folicur in controlling the disease as well as increasing the yield. Other fungicide such as RIL 010/F1 SC, RIL 010/F1 50 SC, Rhizolex 50 WP, Rhizocin 3L, Fulicur (tebuconazole) 250 EW, Contaf 5EC and Tilt (propiconazole) 250 EC were also effective at higher

concentration and was at par with Bavistin (carbendazim) 50 WP. Shield (clopyralid) 2.62 SC proved least effective.

Lore *et al.* (2007) reported that the efficacy of eight fungicides, Tilt 25EC (propiconazole), Bavistin 50 WP (carbendazim), Baycor 25 WP (bitertanol), Contaf 5EC (hexaconazole), Dithane Z-78 (zineb), Kitazin 48 EC (iprobendosol), Saaf 75 WP (carbendazim+mancozeb), and Rhizocin 3L (validamycin), was compared against three economically important diseases of rice namely sheath blight (*R. solani*), sheath rot (*Fusarium moniliforme* [*Gibberella moniliformis*]) and brown spot (*Drechslera oryzae* [*Cochliobolus miyabeanus*]), under laboratory and field conditions (Punjab, India) by artificial inoculation technique during the kharif seasons of 2002, 2003 and 2004. Tilt 25 EC at 0.1% was the most effective fungicide against the three diseases, where the lowest mean disease severity was 15.7% for sheath blight, 6.7% for sheath rot and 9.0% for brown spot, compared to 51.4%, 24.8% and 48.5% in untreated controls of respective diseases. The second best fungicide appeared to be Bavistin 50 WP at 0.1% that was also quite effective against sheath blight (18.1%) and sheath rot (10.5%), followed by Contaf 5EC at 0.1%, but was least effective against brown spot (33.20%). Rhizocin 3 L at 0.25% was also significantly effective against sheath blight. Baycor 25 WP at 0.1% and Dithane Z-78 at 0.25% were the least effective, followed by Kitazin 48EC at 0.2% and Saaf 75 WP at 0.15% against the three diseases. Tilt 25 EC, was also highly effective against grain discoloration, followed by Bavistin 50 WP and Contaf 5 EC. Under laboratory conditions, Tilt 25EC demonstrated a high level of efficacy, followed by Contaf 5EC against the pathogens of the above-mentioned diseases.

Bavistin 50WP was also equally effective against *R. solani* and *F. moniliforme*, but it did not control the growth of *D. oryzae* even at 200 ppm. Rhizocin 3L and Saaf 75WP also showed good activity, but Dithane Z-78 exhibited the least activity against the test pathogens.

Janki-Kandhari (2007) reported that seven fungicides and three botanicals were tested at different concentrations against sheath blight (*Rhizoctonia solani*) of rice cv. Pusa Basmati-1 under field conditions in New Delhi, India, during kharif 2004 and 2005. All the fungicides and botanicals were found effective. However, among fungicides, Armure 30EC (propiconazole + difenoconazole) at 1.0 ml/l recorded the least disease incidence of 31.3%, compared with 77.9% in the untreated control. Among botanicals Achook (azadirachtin) showed the least incidence of 65.1%, compared with 82.9% in the control. Bavistin used as standard fungicide at 1000 ppm was most effective, with disease incidence of 8.4% only.

2.6. Yield loss due to the sheath blight:

Tasi (1975) reported that corresponding yield losses of 43% and 31.9% and 16.3%, occurred for infection at tillering and booting stages in the first and second crops, respectively.

Arunyanart *et al.* (1986) reported that rice plants infected by *Rhizoctonia solani* with disease severity 3, 5, 7 and 9 reduced yield 15, 22, 28 and 40%, respectively.

Correlation between disease severity and percent yield loss, straw weight loss and empty grains was a linear regression.

Shahjahan *et al.* (1986) observed that in a farmer's field trial yield losses were 31.0% and 28.7% at mean disease indices (DI) of 4.8 and 5.0 in susceptible varieties BR2 and BR3, respectively in Bangladesh. The yields were highly and negatively correlated with DI. The regression of DI on yield showed that a reduction of 0.20t ha⁻¹ for BR2 and 0.26 t ha⁻¹ for BR3 might be attributed to per unit increases in DI. Under inoculated condition BR1 and BR5 two susceptible variety had a mean loss of 17.3% and 13.1% with mean DI differences of 4.1(5.4-1.3) and 2.0(4.8-2.8), respectively. Again in separate experiments BR1 had a loss of 14.0% during Aus and 15.4% in IR5 during Aman seasons at DI 5.0 and 4.5, respectively.

Ahn and Mew (1986) studied relation between rice sheath blight and yield. No significant yield reduction was found when RLH was less than 20%. If lesion reached the 3rd sheath from the sheath bearing the flag leaf, and if RLH was about 30% a significant yield loss was seen. A 46% yield loss is possible in milled rice if sheath blight lesion reaches 90% of RLH.

Rajan (1987) reported that percentage yield loss due to sheath blight of rice of 3, 5, and 7 and disease scores in 0-9 scale of SES were 16.8, 22.9, 36.2 and 48.4, respectively.

Naidu (1989) inoculated eight varieties with sheath blight pathogen *R. solani* using a stem tape method and scored for yield. Grain yield losses varied from 49% in Phalguna to 69.1% in Mansui.

Dilla *et al.* (1993) observed that yield per hectare ranged from 3.10 to 7.67 t ha⁻¹ during dry season and relatively lower during the wet season, ranging from 3.46 to 5.96 t ha⁻¹. The effect of sheath blight inoculation was very strong in all cases. The variation in yield losses ranged from 0.27 to 1.29 t ha⁻¹ in dry season and 0.23 to 1.37 t ha⁻¹ in the wet season. Yield loss was strongly affected by amount of inoculums but by population density and nitrogen levels. The value ranged from 3.59 to 27.01% wet season and 3.16 to 28.62% in dry season.

CHAPTER 3

MATERIALS AND METHODS

3. FIELD EXPERIMENT

3. 1. Experimental site

The field experiment was carried out in the field of Plant Pathology Department at Hajee Mohammad Danesh Science and Technology University, Dinajpur during the period of July 2009 to December 2009.

3. 2. Land topography and soil

The place of the field experiment falls under the AEZ-1 (Agro-Ecological-Zone) of Bangladesh and situated at latitude of 25.38 ° N, longitude of 88.41 ° E. The soil type and other characteristics are given below (Anonymous, 1988):

Characteristics of AEZ-1

Name	: Old Himalayan Piedmont Plain
Land type	: Medium high land
Textural class	: Sandy loam
Soil pH	: Above 5.5
Organic matter (%)	: About 1.0%
Field level	: Above flood level

3. 3. Seedling collection

Seedling of rice varieties Hybrid Hera, BRRI Dhan-49 & BR-11 was collected from local farmers of Dinajpur.

3.4. Treatments of the experiment

Three rice varieties viz. Hybrid Hera, BRRI Dhan-4, BR-11, and three fungicides viz. Bavistin 50 WP, Tilt 250 EC, Folicour EW 250 were used in the experiment.

The experiment was conducted with two factors.

Factors were as follows:

i. **Factor A: Varieties**

V_1 = Hybrid Hera

V_2 = BRRI Dhan-49

V_3 = BR-11

ii. **Factor B: Fungicides**

T_0 = Control

T_1 = Bavistin 50 WP (Carbendazim)

T_2 = Tilt 250 EC (Propiconazole)

T_3 = Folicour EW 250 (Tebuconazole)

3.5. Land preparation and fertilizer application

The land was first plowed in late July. Later four-plowings and cross plowings followed by 3 laddering were done during the courses of land preparation. The following doses of manures and fertilizers were applied in the fields as recommended by BARC (Anonymous, 2004).

Manures and fertilizers	Rate / hectare
Poultry Litter	1.50 ton
Urea	150 Kg
TSP	100 Kg
MOP	70 Kg
Gypsum	60 Kg
Zinc Sulphate	10 Kg

The Poultry Litter, TSP, MP, Gypsum, Zinc sulphate were applied at the time of final land preparation. Nitrogen in the form of urea was applied in three equal splits at growth stages 15, 30 and 50 days after transplanting (DAT).

3.6. Transplanting of seedlings in the field

After preparation of land, 25-30 days old seedlings of Hybrid Hera, BRRI Dhan-49 & BR-11 were transplanted in the experimental plots using 1 seedling/hill in Hybrid Hera & 2-3 seedling/hill in BRRI Dhan-49 & BR-11 with a distance of 15cm and 20 cm plant to plant, row to row respectively. Transplanting of seedling was done on 4th August, 2009.

3.7. Intercultural operations

3.7.1. Weeding: Weeding was done in the field as necessary.

3.7.2. Irrigation: Irrigation was given in the field as necessary.

3.8. Preparation and Application of Fungicides

Fungicides were sprayed as solution into the experimental plot except the control plot. Each spray solution was prepared by mixing definite amount of fungicides with tube well water. The whole surface of the plant was sprayed by the solution of the fungicides. In case of control plot plain tube well water was spray on the plants. The fungicides spraying were done in maximum tillering stage of rice plant. Every time the fungicides were freshly prepared prior to application and the spray tank was thoroughly cleaned before filling with the individual spray

material. Special attention was given to complete coverage of the growing plants with the fungicides. Adequate precautions were taken to avoid drifting of spray materials from one plot to the neighboring ones. Details of fungicide used as spray material is given below:

Common Name	Chemical name	Active ingredient	Dose Used (%)
Bavistin 50 WP	Methyl benzimidazol-2-yl-carbamate	Carbendazim (50%)	0.1
Tilt 250EC	1-[2-(2,4-Dichlorophenyl)]-4-propyl-1, (3-Dioxalene-2-yl-methyl)-1 H, 1,2,4-triazole	Propiconazole (25%)	0.1
Folicour EW 250	1-(4-Chlorophenyl)-4,4-dimethyl-3-[1,2,4]triazol-1-ylmethyl-pentan-3-ol	Tebuconazole	0.1

3.9 Design and layout of the experiment:

The experiment was laid out in Randomized Block Design (RCBD) with three replications. The whole plot was divided into three blocks containing 12 plots of 4m x 1.5m size giving total 36 plots. Each of the treatment was set in each block. The space kept between the block was 1.0m and plot to plot was 0.5m.

3.10 Evaluation of Sheath Blight Severity

Ten hills from each unit plot randomly considered for grading the severity of disease on standing plants. Ten selected hills were tagged. Sheath blight severity was recorded in the three growth stages viz. booting, flowering and grain filling. The severity of sheath blight disease was recorded following the Standard Evaluation System for Rice (IRRI, 2001).

The scale was formulated based on percentage of Relative Lesion Height (%RLH), which is given below:

0= Highly resistant = No incidence

1= Resistant = Less than 1% sheath area affected

3= Moderately resistant =1-5% sheath area affected

5= Moderately Susceptible = 6-25% sheath area affected

7= Susceptible = 26-50% sheath area affected

9= Highly Susceptible = 51-100% sheath area affected

The RLH is the average vertical height of uppermost lesion on leaf or sheath expressed as a percentage of the average plant height. The mean value of rating (%RLH) will be determined to get rating score of the material under each treatment. The percent relative lesion height will be calculated as per the formula given below (Ansari, 1995):

$$\% \text{ Relative Lesion Height (RLH)} = \frac{\text{Lesion height}}{\text{Plant height}} \times 100$$

3.11 Data Collection:

Data were collected on following parameters

- Plant height (cm)
- No. of Tiller/Hill
- No. of infected tiller/ hill
- Lesion Height (cm)
- % RLH
- Yield (g)/ 10hill
- Yield (kg)/Plot
- 1000-grain weight (g)
- Total Yield (t/ha)/ plot

3.12 Harvesting, threshing and weighting:

The crop was harvested on 09 December, 2009 at fully ripening stage from field in plot wise. Moreover ten tagged hills of each unit plot were harvested separately. Remaining plants of each plot were also harvested and made bundles. The all bundled were carried to farm office and dried under sunlight for 2 days. The fully dried bundles were threshed carefully and have been taken weight of 1000 seed and yield per plant from 10 tagged hills .Yield per plot was taken adding the seed weight of ten tagged hills to the of rest of the plant in a specific plot .The yield per plot was converted in the yield per plot to the yield per hectare.

3.13 Isolation and Identification of *Rhizoctonia solani*

Disease leaf sheath from the experimental plot was collected by using polyethylene bag was taken to the Plant Pathology laboratory, Hajee Mohammad Danesh Science and Technology University, Dinajpur. The diseased leaf sheath was cut into small pieces (about 0.5-1 cm) from the infected portion along with some healthy portion. The cut pieces were surface sterilize by dipping it in to 10% sodium hypochlorite solution for 2-3 minutes. The pieces were then washed in water three times and were placed on to PDA media in sterilized petridis with help of sterile forceps. The Petridis were then incubated at $25\pm 1^{\circ}\text{C}$ for 7-10 days. Later the pathogen was isolated and purified using hyphal tip culture method and grown on PDA media at $25\pm 1^{\circ}\text{C}$ for 2 weeks and identified according to the key of Kubick and Harman (1998).

3.14 Weather report

The data of monthly average temperature, relative humidity, monthly total rainfall (mm) and sunshine hours received at the experimental site during the period of the study were collected from Wheat Research Institute, Noshipur, Dinajpur. (Appendix II)

3.15 Analysis of the data

Data were subjected to statistical analysis to find out the levels of significance of the experimental results. The mean of all the treatments were calculated and analysis of variance was performed. The significance of differences among pairs of treatment means were evaluated by Duncan's New Multiple Range Test (DMRT) at 5% level of probability (Gomez and Gomez, 1984). All the data were calculated carefully and analyzed with the help of a personal computer in MSTAT-C computer package program.

CHAPTER 4

RESULTS

The effect of different treatments and varieties against sheath blight of rice caused by *Rhizoctonia solani* was evaluated on the basis of plant height (cm), number of tiller/hill, and number of infected tiller/hill, lesion height (cm), percentage relative lesion height (% RLH), yield (g)/10hills, 1000-grain wt (g), and yield (t/ha) presented in this chapter.

Symptoms:

The lesions were usually observed on the leaf sheaths although leaf blades. The initial lesions were small, ellipsoid or ovoid, and greenish-gray and usually developed near the water line in the field. Under favorable conditions, lesions were enlarged and coalesced forming bigger cobra skin like lesions with irregular outlined and grayish-white center with dark brown borders (Plate 1). The presence of several large spots on a leaf sheath causes the death of the whole leaf. A pure culture of *Rhizoctonia solani* was prepared on PDA (Plate-2) from infected leaf sheath of rice.



Cobra skin like
lesion develop

Plate 1. A typical symptom of sheath blight disease of rice

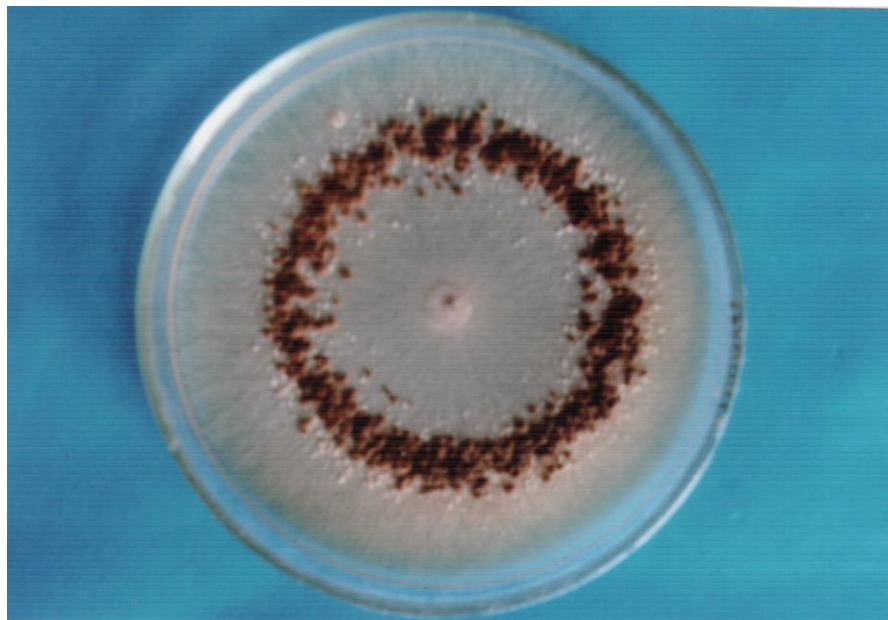


Plate 2. Pure culture of *Rhizoctonia solani* showing sclerotia on PDA



Plate 3. Control (non sprayed) plot



Plate 4. Folicour EW 250 three sprayed plot

4.1. Effect of fungicides on morphological characters, yield and disease parameters of rice plant

4.1.1. Effect of fungicides on plant height (cm)

Mean Plant height was significantly influenced by the fungicide application which is presented in table 1. Mean plant heights were recorded in different fungicides ranged from 108.40cm to 112.50cm. The tallest plants (112.50cm) were recorded in Tilt 250 EC sprayed plot where the shortest plants (108.40cm) were obtained from the control plot.

4.1.2. Effect of fungicides on numbers of tiller/hill

The effect of different fungicides on number of tiller/hill of rice were determined and presented in Table 1. The number of tillers per hill was significantly different in respect of different fungicides used. The total numbers of tiller/hill were 11.32, 12.85, 13.10 and 13.65 in Control, Bavistin 50 WP, Tilt 250 EC and Folicour EW 250 treatment, respectively. The highest numbers of tiller/hill (13.68) were recorded in the Folicour EW 250 sprayed plot. The lowest numbers of tiller/hill (11.32) were recorded in the Control plot.

4.1.3. Effect of fungicides on numbers of infected tiller/hill

It was observed that the fungicides showed significant effect on number of infected tiller per hill (Table 1). Number of infected tiller per hill ranged from 4.92 to 5.28 under field condition. The number of infected tiller/hill was 5.28, 5.06, 5.09 and 4.92 in Control, Bavistin 50 WP, Tilt 250 EC and Folicour EW 250

treatment, respectively. The highest numbers of infected tiller/hill (5.28) were recorded in the Control plot. The lowest numbers of infected tiller/hill (4.92) were recorded in the Folicour EW 250 sprayed plot. Bavistin 50 WP and Tilt 250 EC treatment were significantly different from Control or Folicour EW 250. The numbers of infected tiller per hill decrease (6.81%) over control were recorded in the fungicides Folicour EW 250 (Fig. 1)

4.1.4. Effect of fungicides on lesion height (cm)

The fungicides differed significantly in respect of lesion height per infected plant (Table 1). Lesion heights per infected plant under different fungicides were 4.69, 4.13, 3.82 and 3.23 in Control, Bavistin 50 WP, Tilt 250 EC and Folicour EW 250 treatments, respectively. The highest lesion height (4.69) was observed in Control and lowest lesion height (3.23) was observed in Folicour EW 250 sprayed plot. Bavistin 50 WP and Tilt 250 EC fungicides were statistically similar but different from Control and Folicour EW 250.

4.1.5. Effect of fungicides on % relative lesion height (%RLH)

The effect of different fungicides on relative lesion height (%) is shown in Table 1. Percent relative lesion height was 4.16, 3.62, 3.98 and 3.42 in Control, Bavistin 50 WP, Tilt 250 EC and Folicour EW 250, respectively. The highest % relative lesion height (4.16) was recorded in the Control plot. The lowest % relative lesion height (3.42) was recorded in the Folicour EW 250 sprayed plot. Bavistin 50 WP and Tilt 250 EC fungicides were significantly different from Control and Folicour

EW 250. The % relative lesion height decrease (17.78%) over control was recorded in the fungicides Folicour EW 250 (Fig. 2).

4.1.6. Effect of fungicides on yield (g)/10hills

The yield (g)/10hills differed significantly from one fungicide to another (Table 1). The yield (g)/10hills ranged from 220.20 to 270.30 where the highest and lowest yield was recorded in Folicour EW 250 and Control plot respectively. Control and Tilt 250 EC fungicides were significantly different from Bavistin 50 WP and Folicour EW 250. The highest yield (g)/10hills increased (22.75%) over control were recorded in the fungicide Folicour EW 250 (Fig. 3).

4.1.7. Effect of fungicides on 1000- grain weight (g)

No significant variation was found among the fungicides (Table 1) on 1000- grain weight (g). But the highest 1000- grain weight (g) was recorded 25.08 in Folicour EW 250 sprayed plot and the lowest (24.50) in Tilt 250 EC sprayed plot.

4.1.8 Effect of fungicides on yield (t/ha)

The effect of fungicides differ significantly in respect of yield (t/ha) (Table 1). The yield (t/ha) ranged from 4.24 to 5.81. The highest yield (t/ha) (5.81) were recorded in the Folicour EW 250 sprayed plot. The lowest yield (t/ha) (4.24) were recorded in the Control plot. Bavistin 50 WP and Tilt 250 EC fungicides were statistically similar but different from Control and Folicour EW 250. The highest yield (t/ha) increased (37.02%) over control were recorded in the fungicide Folicour EW 250 (Fig. 4).

Table 1. Effect of fungicides on morphological characters, yield and disease parameters of rice plant

Fungicides	Plant height (cm)	No. of tiller/hill	No. of infected tiller/ hill	Lesion height (cm)	% RLH	Yield (g)/ 10hills	1000-grain wt (g)	Yield (t/ha)
Control	108.40 c	11.32 c	5.28 a	4.69 a	4.16 a	220.20 b	24.88 a	4.24 c
Bavistin 50WP	111.40 ab	12.85 b	5.06 b	4.13 b	3.62 c	246.00 ab	25.00 a	5.02 b
Tilt 250 EC	112.50 a	13.10 ab	5.09 b	3.82 b	3.98 b	234.70 b	24.50 a	5.01 b
Folicour EW 250	110.70 b	13.68 a	4.92 c	3.23 c	3.42 d	270.30 a	25.08 a	5.81 a
LSD(0.05)	1.39	0.77	0.16	0.52	0.15	31.80	0.74	0.78
CV %	1.29	6.12	3.22	11.71	3.69	13.40	3.06	14.36

In a column, figures having similar letter(s) do not differ significantly where as figure s bearing dissimilar letter(s) differ significantly(as per DMRT).

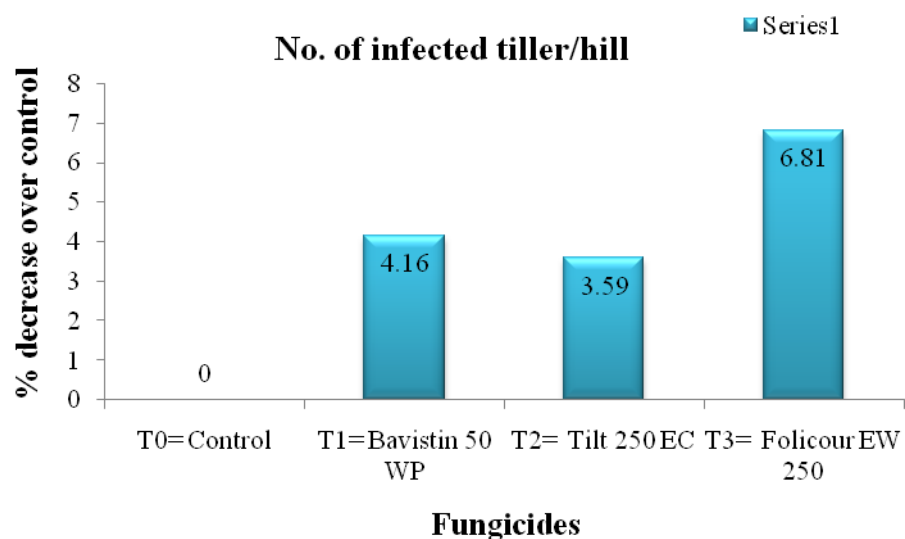


Fig.1. Percent decrease number of infected tiller per hill over control due to fungicidal application

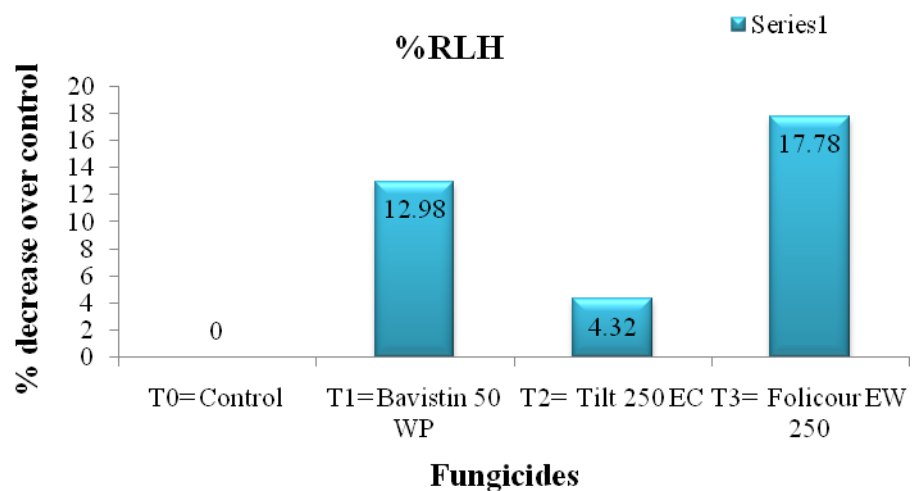


Fig.2. Percent decrease %RLH over control due to fungicidal application

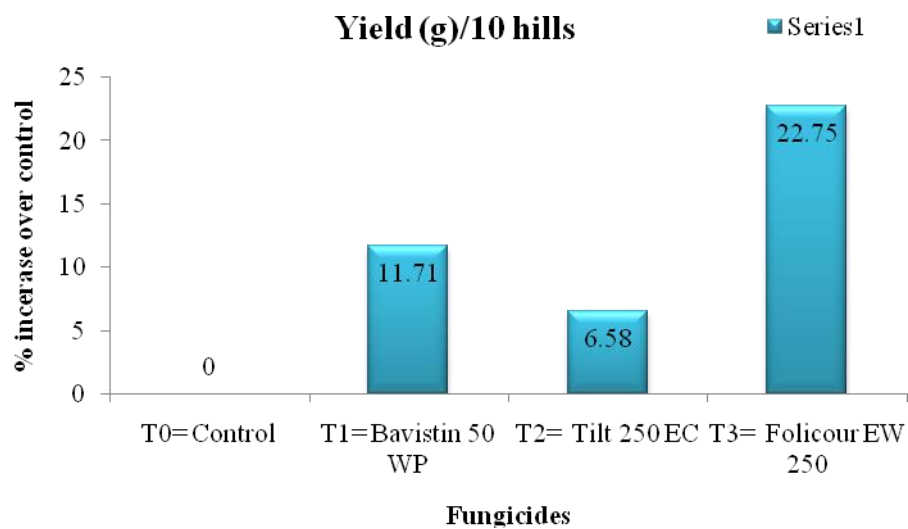


Fig.3. Percent increase yield (g)/10hills over control due to fungicidal application

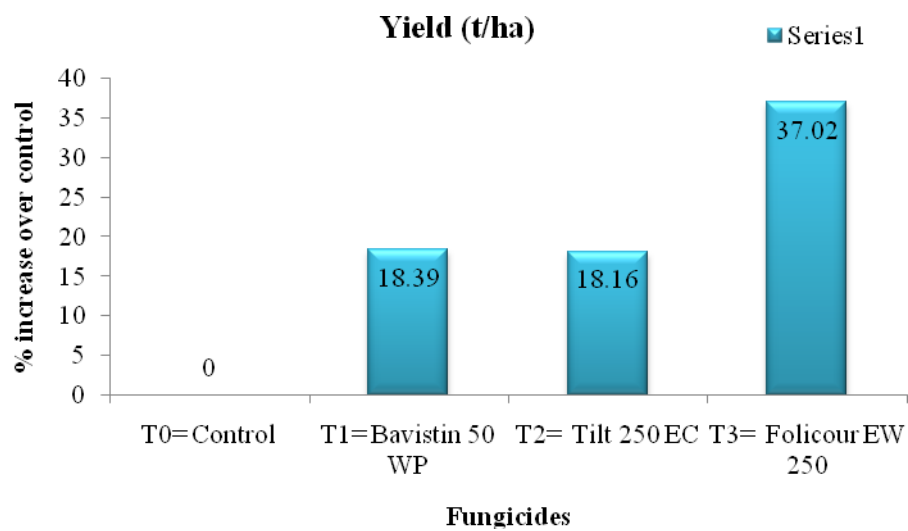


Fig.4. Percent increase yield (t/ha) over control due to fungicidal application

4.2. Effect of varieties on morphological character, yield and disease parameters of rice plant

4.2.1. Effect of varieties on plant height (cm)

The plant height was significantly difference from variety to variety (Table 2). The plant height was found 116.80, 110.10 and 106.00 cm in Hybrid Hera, BRRI Dhan-49 and BR-11 variety, respectively. The tallest plants heights (116.80 cm) were recorded in Hybrid Hera variety. The shortest plants heights (106.00 cm) were obtained from the BR-11 variety.

4.2.2. Effect of varieties on number of tiller/hill

It was observed that the varieties showed significant effect on numbed of tiller per hill (Table 2). Number of tiller per hill ranged from 9.73 to 16.65 under field condition. The number of tiller/hill was 9.73, 12.72 and 16.65 in Hybrid Hera, BRRI Dhan-49 and BR-11 variety, respectively. The highest numbers of tiller/hill (16.65) were recorded in the BR-11 variety. The lowest numbers of tiller/hill (9.73) were recorded in the Hybrid Hera variety.

4.2.3. Effect of varieties on number of infected tiller/hill

The effect of different varieties on number of infected tiller/hill of rice were determined and presented in Table 2. The number of infected tillers pre hill significantly different in respect of different variety used. The number of infected tiller/hill was 3.31, 4.22 and 8.31 in Hybrid Hera, BRRI Dhan-49 and BR-11 variety, respectively. The highest numbers of infected tiller/hill (8.31) were recorded in the BR-11 variety. The lowest numbers of infected tiller/hill (3.31) were recorded in the Hybrid Hera variety.

4.2.4. Effect of varieties on lesion height (cm)

The effects of different varieties on lesion height (cm) were found to be significant (Table 2). Lesion heights were 4.16, 4.46 and 5.11 in Hybrid Hera, BRRI Dhan-49 and BR-11 variety, respectively. The highest lesion height (5.11) was recorded in the BR-11 variety. The lowest lesion height (4.16) was recorded in the Hybrid Hera variety. Hybrid Hera and BRRI Dhan-49 variety were significantly different from BR-11 variety.

4.2.5. Effect of varieties on % relative lesion height (% RLH)

The varieties were differed significantly in respect of % relative lesion height per infected plant (Table 2). Percent relative lesion heights per infected plant under different varieties were 3.72, 4.05 and 4.82 in Hybrid Hera, BRRI Dhan-49 and BR-11 varieties, respectively. The highest % relative lesion height (4.82) was observed in BR-11 variety and lowest % relative lesion height (3.72) was observed in Hybrid Hera variety.

4.2.6. Effect of varieties on yield (g)/10hills

The effect of different varieties on yield/10hills of rice were determined and presented in Table 2. The yield/10hills were significantly different in respect of different variety used. The yield/10 hills were 222.20, 180.50 and 325.70g in Hybrid Hera, BRRI Dhan-49 and BR-11 variety, respectively. The highest yield/10hills (325.70 g) were recorded in the BR-11 variety. The lowest yield/10hills (180.50 g) were recorded in the BRRI Dhan-49 variety.

4.2.7. Effect of variety on 1000-grain weight (g)

The effects of different varieties on 1000-grain weight (g) were found to be significant (Table 2). 1000-grain weight (g) was 25.64, 20.04 and 28.91 in Hybrid Hera, BRRI Dhan-49 and BR-11 variety, respectively. The highest 1000-grain

weight (g) (28.91) was recorded in the BR-11 variety. The lowest 1000- grain weight (g) (20.04) was recorded in the variety BRRI Dhan-49.

4.2.8. Effect of varieties on yield (t/ha)

The yield (t/ha) was significantly difference from variety to variety (Table 2). The yield was found 5.10, 5.65 and 5.93 t/ha in Hybrid Hera, BRRI Dhan-49 and BR-11 variety, respectively. The highest yields (5.93 t/ha) were recorded in BR-11 variety. The lowest yields (5.10 t/ha) were obtained from the Hybrid Hera variety. And intermediate result found in BRRI Dhan-49 variety.

Table 2. Effect of varieties on morphological characters, yield and disease parameters of rice plant

Variety	Plant height (cm)	No. of tiller/ hill	No. of infected tiller/ hill	Lesion height (cm)	%RLH	Yield (g)/ 10hills	1000-grain wt (g)	Yield (t/ha)
Hybrid Hera	116.80 a	9.73 c	3.31 c	4.16 b	3.72 c	222.20 b	25.64 b	5.10 b
BRRI Dhan-49	110.10 b	12.72 b	4.22 b	4.46 b	4.05 b	180.50 c	20.04 c	5.65 ab
BR-11	106.00 c	16.65 a	8.31 a	5.11 a	4.82 a	325.70 a	28.91 a	5.93 a
LSD (0.05)	1.21	0.67	0.14	0.45	0.13	27.54	0.64	0.67
CV %	1.29	6.12	3.22	11.71	3.69	13.40	3.06	14.36

In a column, figures having similar letter(s) do not differ significantly where as figure s bearing dissimilar letter(s) differ significantly(as per DMRT).

4.3. Interaction effect of fungicides and varieties on morphology, yield and disease parameters of rice plant

The interaction effects of fungicides and varieties on the morphology, yield and disease parameters of rice plant like plant height (cm), numbers of tiller/hill, numbers of infected tiller/hill, lesion height (cm), % relative lesion height (% RLH) yield (g)/10hills, 1000-grain wt (g), yield (t/ha) were found significant (Table 3).

Yield of rice was significantly influenced by different treatment combination. Among the treatment combination BRRI Dhan-49 along with Folicour EW 250 sprayed plot gave the highest yield (6.27 t/ha) compared with other treatment combination. Lowest yield (4.64 t/ha) was recorded in Hybrid Hera along with Control plot. In case of yield (g)/10hills, highest yield (366.20 g) was found in treatment combination BRRI Dhan-49 along with Folicour EW 250 sprayed plot and lowest yield (165.5 g) was found in Hybrid Hera along with Control plot (Table 3).

In case of disease parameters, numbers of infected tiller/hill significantly influenced by different treatment combination. Among the treatment combination BR-11 along with control gave the highest numbers of infected tiller/hill (8.52) compared with other treatment combination. And lowest numbers of infected tiller/hill (3.00) was recorded in Hybrid Hera along with Folicour EW 250 (Table 3).

On the other hand, percent relative lesion height (% RLH) was also significant influence by different treatment combination. Among the treatment combination BR-11 along with control gave the highest % relative lesion height (5.08%) compared with other treatment combination. And lowest % relative lesion height (3.53%) was recorded in Hybrid Hera along with Folicour EW 250 sprayed plot (Table 3).

Table 3. Interaction effects of the fungicides and varieties on morphological characters, yield and disease parameters of rice plant

Treatment Combination		Plant height (cm)	No. of tiller/hill	No. of infected tiller/hill	Lesion height (cm)	%RLH	Yield (g) /10hills	1000-seed wt (g)	Total yield (t/ha)
Variety	Fungicide								
Hybrid	Control	115.10 b	9.08 d	8.24 ab	4.66 a	4.71 b	165.50 e	20.13 d	4.64 b
	Bavistin 50 WP	108.50 def	12.25 c	4.26 c	4.65 a	4.28 c	203.70 cde	19.80 d	5.08 ab
	Tilt 250 EC	104.30 h	16.19 a	3.35 d	5.29 a	4.08 cd	291.50 b	25.40 c	5.61 ab
	Folicour EW 250	116.70 ab	9.60 d	3.00 e	4.44 a	3.53 f	217.80 cde	28.97 ab	5.53 ab
BRRI Dhan-49	Control	109.80 cde	13.00 bc	8.35 ab	4.48 a	4.88 ab	171.30 de	20.13 d	5.78 ab
	Bavistin 50 WP	105.60 gh	16.71 a	4.19 c	5.15 a	4.07 cd	176.60 cde	25.90 c	6.02 ab
	Tilt 250 EC	117.30 ab	9.82 d	3.23 d	4.33 a	3.69 ef	231.30 cd	27.97 b	5.07 ab
	Folicour EW 250	110.50 cd	11.96 c	4.23 c	4.42 a	4.00 d	366.20 a	29.43 a	6.27 a
BR-11	Control	106.50 fgh	16.76 a	8.52 a	5.02 a	5.08 a	296.20 b	25.40 c	5.07 ab
	Bavistin 50 WP	118.20 a	10.42 d	3.37 d	3.23 b	3.58 ef	236.10 c	29.27 ab	5.17 ab
	Tilt 250 EC	111.70 c	13.68 b	4.19 c	4.29 a	3.84 de	208.70 cde	20.10 d	5.49 ab
	Folicour EW 250	107.70 efg	16.94 a	4.14 c	4.98 a	4.62 b	348.90 ab	25.87 c	5.98 ab
LSD (0.05)		2.42	1.35	0.28	0.90	0.26	55.08	1.28	1.35
CV		1.29	6.12	3.22	11.71	3.69	13.40	3.06	14.36

In a column, figures having similar letter(s) do not differ significantly where as figure s bearing dissimilar letter(s) differ significantly(as per DMRT).

4.4. Correlation and Regression Study:

4.4.1. Relationship between lesion height (cm) and yield (gm)/10hills

The negative but significant correlation was found between lesion height (cm) and yield (g)/10hills in treatments having ($r = - 0.81^{**}$) and the linear correlation could be describe by $Y = 367.28 - 31.36x$. (Fig.5). It is indicates that yield (g)/10hills was declined with the increasing lesion height.

4.4.2. Relationship between %RLH and yield (g)/10hills

In respect of %RLH and yield (g)/10hills, there appeared negative correlation between them having ($r = - 0.93^{**}$) and it followed the regression equation $Y = 473.64 - 60.79x$ (Fig. 6). The result indicate that there was an increasing of %RLH with the decreasing of yield (g)/10hills.

4.4.3. Relationship between number of infected tiller/hill and yield (g)/10hills

Significant and negative correlation was found in number of infected tiller/hill and yield (g)/10hills for the treatments having ($r = - 0.92^{**}$) and followed the regression equation $Y = 940.76 - 137.10x$ (Fig. 7). The result denoted that yield (g)/10hills was reduced with the increasing number of infected tiller/hill.

4.4.4. Relationship between plant height (cm) and yield (g)/10hills

The correlation between plant height and yield (g)/10hills was negative ($r = - 0.99^{**}$) with the treatments and the relation could be described by $Y = 1590 - 12.16x$ (Fig.8). The result depicted that there was an increasing of plant height with the decreasing of yield (g)/10hills.

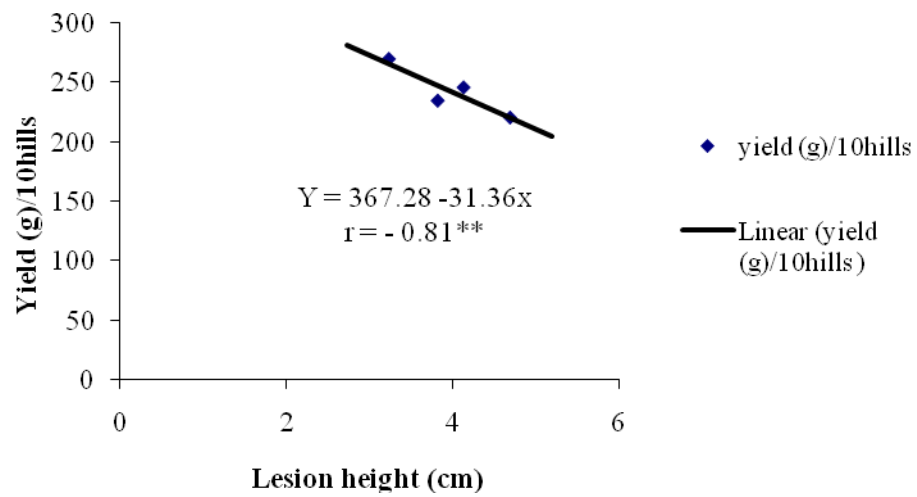


Fig.5. Relationship between lesion height (cm) and yield (g)/10hills

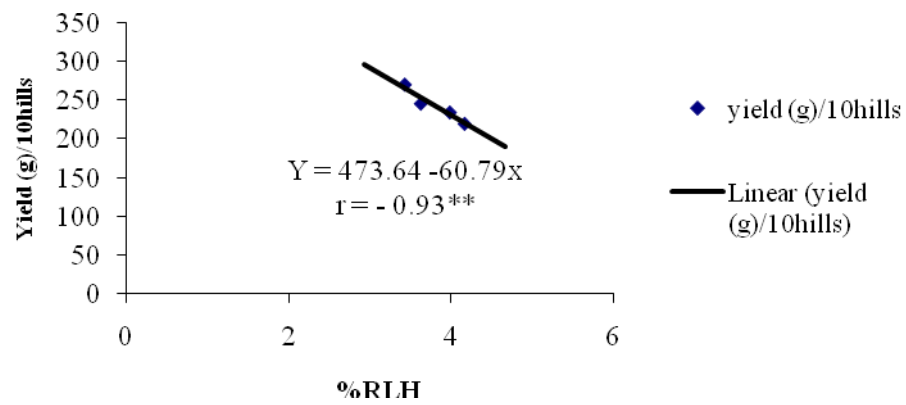


Fig.6. Relationship between %RLH and yield (g)/10hills

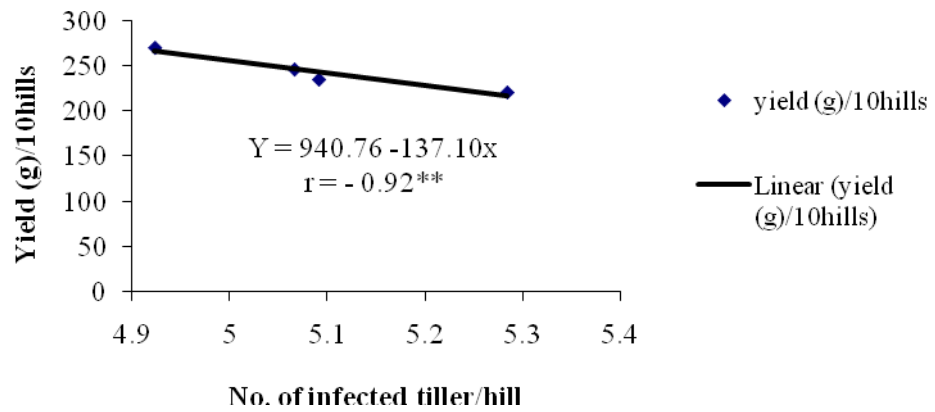


Fig.7. Relationship between number of infected tiller/hill and yield (g)/10hills

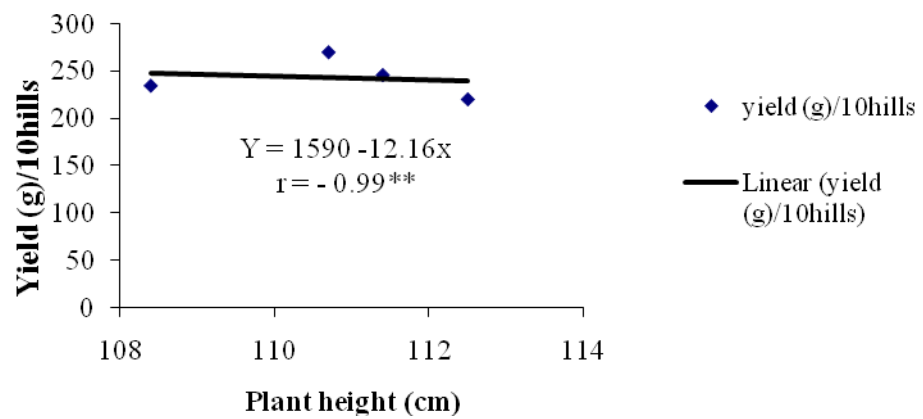


Fig.8. Relationship between plant height (cm) and yield (g)/10hills

CHAPTER 5

DISCUSSION

The sheath blight of rice caused by *Rhizoctonia solani* is documented to be major and destructive disease occurring in Bangladesh specially of modern high input requiring cultivars (Miah *et al.*, 1985; Shahjahan *et al.*, 1987). The present study was carried out with four treatments viz Control, Bavistin 50 WP, Tilt 250 EC and Folicour EW 250 for the management of sheath blight of three rice varieties such as, Hybrid Hera, BRRI Dhan-49 and BR-11 caused by *Rhizoctonia solani* under field condition. From the present study it is evident that different treatments resulted reduction of number of infected tiller of rice in comparison with control. Sprays of Folicour EW 250 @ 0.1% resulted minimum number of infected tillers and maximum number were found in Control under field condition but the highest numbers of infected tiller/hill (8.31) were recorded in the variety BR-11 and lowest numbers of infected tiller/hill (3.318) were recorded in the Hybrid Hera variety. Jones *et al.* (1987) reported that Tilt (Propiconazole) applied twice significantly reduced sheath blight disease severity and increased yield. Groth *et al.* (1993) also reported that Tilt (Propiconazole) gave good control of sheath blight disease.

Regarding the effect of fungicidal sprays on the plant height marked difference has been observed. Comparatively the tallest plants (112.50cm) were recorded in treatment Tilt 250 EC sprayed plot and shortest plants (108.40cm) were obtained from the Control treatment but in case of varieties, tallest plants heights

(116.80 cm) were recorded in hybrid variety and shortest plants heights (106.00 cm) were obtained from the BR-11 variety. The normal luxuriant leafy growth of the high yielding short varieties seems to produce a dense canopy more favourable disease encouraging micro climate and also for successful growth of fungus on leaf sheaths. In the taller plants with comparatively early leaf senescence, the lower leaf dry up, some of which fall off as the plant grows and new leaves emerge on the upper nodes far above the plant base. So, the lower part of the culm remains clear partly covered with dead and dry leaf sheaths. This condition in taller plant allows more sun rays and better ventilation than the shorter plants. Thus the micro climate as well as the lack of living green leaf sheath at the infection site remain unfavourable for the successful disease development. The sheath blight development in short/semi-dwarf line is more because of shorter distance between the sheath position at water line, the usual infection court, and the flag leaf or the panicals. These are may be main reasons why relatively taller plants produced with the help of Tilt 250 EC to reduce the growth of pathogen which remain erect show more resistance to this disease.

The sclerotia of *R. solani* develop the blight symptoms on the sheath and destroyed the chlorophyllous tissue. It is a soil-borne pathogen and initial infection develop on sheath of the plants. The height of lesion can be changed as per with the growth of plant. Folicur EW 250 is a systemic fungicide and helps to reduce disease severity in the field conditions. The highest lesion height (5.11) was recorded in the variety BR-11 and lowest lesion height (4.16)

was recorded in the Hybrid Hera variety. According to Shahjahan *et al.* (1990) the sclerotia production in the fungicides sprayed plots was less coinciding with the reduced development in those plots. Belmar *et al.* (1987) observed a linear correlation between the number of *R. solani* sclerotia from soil and sheath blight incidence at the panical initiation growth stage. Thus, the treatments have a significant role in reducing the inoculum density of the respective plot and thereby minimize the chance of becoming severe havoc of sheath blight during next seasons.

From the present study it was evident that different treatments resulted in reduction of percent relative lesion height (% RLH) in comparison to control. The % RLH was found significantly indeferent with some extent. Comparatively, the lower % RLH were observed in case of Folicour EW 250 and the highest % RLH was found in control. Mia (1999) also recommended Tilt 250 EC at the rate of 1L/ha for spraying at PI to boot stage for successful control of sheath blight disease of rice.

The effect of treatments on yield performance was significantly differed except 1000-grain weight. The 1000-grain wt of all treatments ranged 24.50 to 25.08 g. Arunyanart and Surin (1984) also observed significant difference among the 1000 grain weight, where highest weight was found from disease free plant.

Grain yield of rice profoundly varied from within treatment where the yield was 4.24 to 5.81 t/ha. The highest yield was recorded by Folicour EW 250

spread plot of BR-11 rice. The lowest yield was recorded in Control and Hybrid Hera. Hybrid Hera rice variety produce comparatively low yield due to low genetically and environmental potentiality. Sharma (1985) reported that 30.34 and 22.88 % yield loss occurred in cv. BR1 and BR 6, respectively due to the sheath blight.

Among the treatment combination BRRI Dhan-49 along with Folicour EW 250 gave the highest yield (6.27 t/ha) and lowest yield (4.64 t/ha) was recorded in Hybrid along with Control. On the other hand, highest yield/10hills (366.20 g) was found in treatment combination BRRI Dhan-49 along with Folicour EW 250 and lowest yield/10 hill (165.50 g) was found in Hybrid Hera along with Control.

Among the treatment combination BR-11 along with control gave the highest number of infected tiller/hill (8.52) compared with other treatment combination and lowest number of infected tiller/hill (3.00) was recorded in Hybrid Hera along with Folicour EW 250. On the other hand, percent relative lesion height (% RLH) was also significant influence by different treatment combination. BR-11 along with control gave the highest percent relative lesion height (5.03%) compared with other treatment combination and lowest percent relative lesion height (3.53%) was recorded in hybrid along with Folicour EW 250.

Considering the over all performances of treatments applied in management of sheath blight of rice, the best performance was shown in Folicour EW 250

followed by Bavistin 50 WP and Tilt 250 EC but in case of varieties yield performance was good in BR-11 and diseases severity was low in Hybrid Hera Varsity.

CHAPTER 6

SUMMARY AND CONCLUSION

The field experiment was carried out in the field of Plant Pathology Department, Hajee Mohammad Danesh Science & Technology University, Dinajpur during the period of July 2009 to December 2009 to find out the prevalence and fungicidal management of sheath blight of three rice varieties. Four treatments viz. Bavistin 50 WP, Tilt 250 EC and Folicour EW 250 with Control were evaluated against *R. solani* causing sheath blight of rice on three varieties viz. Hybrid Hera, BRRI Dhan-49, BR-11. The experiments were laid out in two factors Randomized Complete Block Design (RCBD) with three replications. Data were collected on disease incidence and yield contributing characters.

The treatments and varieties had remarkable different on plant height. The tallest plants (112.50cm) were recorded in treatment Tilt 250 EC sprayed plot. Where shortest plants (108.40cm) were recorded in the Control plot. The tallest plants height (116.80 cm) was recorded in Hybrid Hera variety and the shortest plants height (106.00 cm) was obtained from the BR-11 variety. The highest numbers of tiller/ hill (13.68) were recorded in the Folicour EW 250 sprayed plot and lowest numbers of tiller/hill (11.32) were recorded in the Control plot. In case of varieties the highest numbers of tiller/hill (16.65) were recorded in the BR-11 variety. The lowest numbers of tiller/hill (9.73) were recorded in the Hybrid Hera variety.

Regarding number of infected tillers per hill, the highest numbers of infected tiller/hill (5.28) were recorded in the Control plot. The lowest numbers of infected tiller/hill (4.92) were recorded in the Folicour EW 250 sprayed plot. The highest numbers of infected tiller/hill (8.31) were recorded in the BR-11 variety. The lowest numbers of infected tiller/hill (3.31) were recorded in the Hybrid Hera variety. Among the treatment combination BR-11 along with control gave the highest number of infected tiller/hill (8.52) compared with other treatment combination. And lowest number of infected tiller/hill (3.00) was recorded in Hybrid Hera along with Folicour EW 250 sprayed plot.

Lesion heights per infected plant under different treatments were varied from 3.43 to 4.69. The highest lesion height (4.69) was observed in Control plot and lowest lesion height (3.43) was observed in Folicour EW 250 sprayed plot. But in case of varieties the highest lesion height (5.11) was recorded in the y BR-11variet. The lowest lesion height (4.16) was recorded in the Hybrid Hera variety. The highest % relative lesion heights (4.16) were recorded in the Control plot. The lowest % relative lesion heights (3.42) were recorded in the Folicour EW 250 sprayed plot. Beside the varieties the highest % relative lesion height (4.82) was observed in BR-11 and lowest % relative lesion height (3.72) was observed in Hybrid Hera variety. Among the treatment combination BR-11 along with control gave the highest % relative lesion height (5.08%) compared with other treatment combination. And lowest % relative lesion height (3.53%) was recorded in Hybrid Hera along with Folicour EW 250

sprayed plot. The % relative lesion height decrease (17.78%) over control was recorded in the treatment Folicour EW 250.

The effects of different fungicides and varieties on yield (g)/10hills were found to be significant. The yield (g)/10hills ranged from 220.20 to 270.30g where the highest and lowest yields were made in Folicour EW 250 and Control plot respectively. The highest yield (g)/10hills increased (22.75%) over control were recorded in the fungicide Folicour EW 250. Among the varieties the highest yield/10 hills (325.70 g) were recorded in the BR-11 variety and the lowest yield/10 hills (180.50 g) were recorded in the BRRI Dhan-49 variety. On the other hand, highest yield/10 hill (336.20 g) was found in treatment combination BR-11 along with Folicour EW 250 sprayed plot and lowest yield/10 hill (165.50 g) was found in Hybrid Hera along with Control plot. The highest yield (t/ha) (5.81) were recorded in the Folicour EW 250 sprayed plot and lowest yield (t/ha) (4.24) were recorded in the Control plot. The highest yield (t/ha) increased (37.02%) over control were recorded in the fungicide Folicour EW 250. In case of varieties the highest yields (5.93 t/ha) were recorded in BR-11 Variety and the lowest yields (5.10 t/ha) were obtained from the Hybrid Hera variety. And intermediate result found in BRRI Dhan-49 variety. Among the treatment combination BRRI Dhan-49 along with Folicour EW 250 sprayed plot gave the highest yield (6.27 t/ha) compared with other treatment combination. And lowest yield (4.64 t/ha) was recorded in Hybrid Hera along with Control plot.

From the present findings it may be concluded that

1. BR-11 was the best variety among the three varieties in aspect of yield and Hybrid Hera showed lowest disease incidence.
2. Folicour EW 250 was a promising fungicide for lowering sheath blight incidence of rice.

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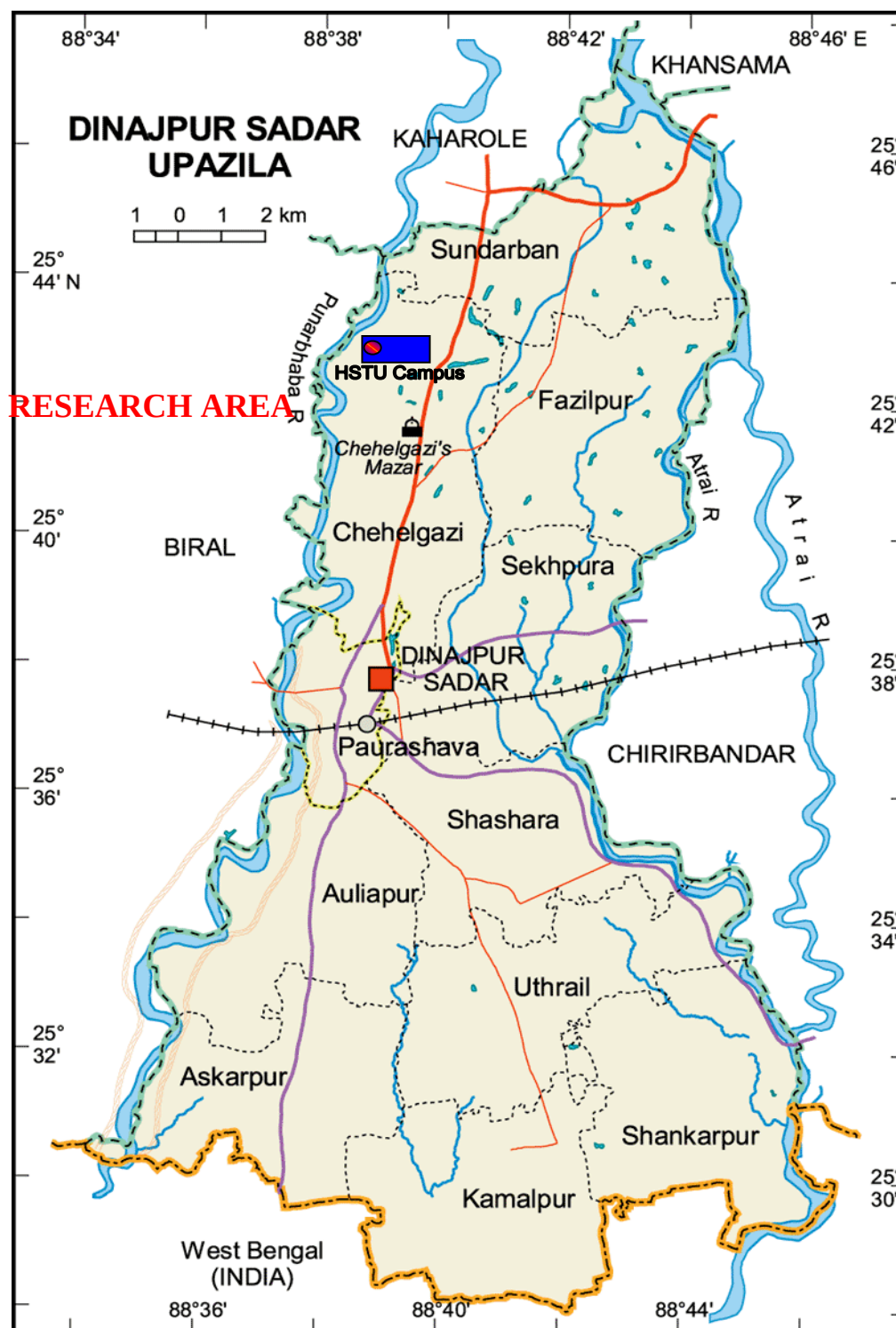
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APPENDICES

Appendix I. Location of the experimental site (map of Dinajpur Sadar Upazila showing the research plot).



Appendix II. Weather data of the experimental site during the period from July to December, 2009

Month	** Air Temperature (C)			**Rainfall (mm) (Minimum)	**Relative Humidity (%)	*Sunshine (hrs)	** Wind speed (mm)
	Maximum	Minimum	Average				
July	31.72	26.54	29.13	----	85.58	3.57	-----
August	31.64	26.32	28.98	128	88.35	4.37	1.16
September	30.54	25.43	27.99	254	89.93	3.45	1.12
October	31.62	23.86	27.74	370	86.10	3.31	1.69
November	29.70	19.09	24.40	00	82.37	6.63	0.39
December	26.71	13.43	20.07	00	77.00	7.93	-----

* = Monthly total

** = Monthly average

Source: Meteorological Station, Wheat Research Center, Nosipur,
Dinajpur (2009).

Appendix II. Experimental Lay out

Plot Size: 1.5m X 4m

Plot to plot Distance: 0.5 m

Replication to Replication distance: 1m

South	West						North		
	R ₁			R ₂				R ₃	
	Hybrid								
	T ₀		1m	T ₃		1m		T ₂	
	T ₁			T ₂				T ₀	
	T ₃			T ₀				T ₃	
	T ₂			T ₁				T ₁	
	BRRI Dhan-49								
	T ₃		1m	T ₂		1m		T ₀	
	T ₂			T ₀				T ₁	
	T ₀			T ₃				T ₃	
	T ₁			T ₁				T ₂	
	BR-11								
	T ₂		1m	T ₀		1m		T ₃	
	T ₀			T ₁				T ₂	
	T ₃			T ₃				T ₀	
T ₁		T ₂		T ₁					

East

T₀=Control

T₁= Bavistin 50 WP (Carbendazim)

T₂= Tilt 250 EC (Propiconazole)

T₃= Folicour EW 250 (Tebuconazole)